Comparative Evaluation of Text Summarization and Question Answering Models with RAG and LLMs

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***Abstract*—The continuous expansion of digital information requires essential automated text summarization and question-answering (QA) systems. This research assesses how Retrieval-Augmented Generation (RAG) systems perform alongside different NLP models when conducting summarization and generating questions and answering queries. This research analyzes benchmark datasets with standard evaluation metrics including ROUGE, BLEU, F1, etc. to determine which models achieve the best balance of accuracy and efficiency within an NLP pipeline. The research establishes progress in developing smart information search systems.**

I. Introduction and Statement of the Problem

The rapidly expanding volume of online content has produced crucial barriers against effective retrieval and understanding of relevant information. The dependency on automated text processing tools by individuals and organizations demands that Natural Language Processing (NLP) techniques become even more essential. Text summarization together with question-answering (QA) systems represent two leading NLP methods that extract important insights from text while generating summaries. The joint application of these tools produces uncertain results regarding maintainable coherence alongside accurate contexts plus proper response delivery. The inconsistencies appear more pronounced in situations where one model's output functions as an input to additional models including question generation or answering systems.

This study evaluates an NLP pipeline integrating 20 summarization models, 3 question generators, and 2 QA systems within a RAG framework. It compares individual and combined model performance, revealing strengths, weaknesses, and synergies. The findings offer insights to improve end-to-end NLP workflows and guide future automated system development.

II. Review of Literature

Sasidhar and Christy (2024) developed an innovative system which brings together GPT-4 with question-answering (QA) and text summarization pipelines. Their research demonstrated real-time processing's vital role when dealing with media file transfers. The researchers showed how BERT and DistilBERT transformer models enhance the retrieval system through their implementation of combined summarization and question-answering functions within one framework. The research confirms that large language models possess the ability to control diverse data streams while producing context-based responses quickly in real-time operational environments.

The authors conducted an additional study which used automated NLP technology exclusively for educational content analysis. Their research investigated summarize-and-generate methods to streamline the information retrieval process from instructional materials. These findings demonstrated how automated approaches transform educational technologies into systems capable of delivering adaptive interactive educational experiences. The research findings demonstrated the requirement of connecting summary sections with appropriate contextual questions because such capabilities form the foundation of superior intelligent tutoring systems alongside content delivery systems.

Bohnet et al. (2024) implemented a side-by-side comparative assessment as part of their detailed evaluation process for QA systems. The research team studied long-context models because these models excel at processing lengthy text sequences to preserve full contextual information. Automatic question generation worked alongside QA evaluation to build a complete scoring and benchmarking system for the researchers. The evaluation method enabled researchers to determine which QA systems performed optimally under different conditions for staying relevant to context while de-

livering accurate answers to long queries. The research group makes valuable progress toward creating interpretable scalable frameworks for processed language tasks.

III. Objectives of the Study

The primary objectives of this project include:

- The research examines 20 text summarization algorithms to produce accurate, informative summaries while keeping the summaries direct and short using various benchmark datasets.
- Web-based research aims to determine the capability of two QA systems as they answer questions developed from summary text.
- This analysis examines three question generation models to assess the quality and relevance of their generated questions.

IV. Data Collection

The dataset employed for this research is "arxiv-metadata-oai-snapshot.json" sourced from Kaggle. It contains metadata for over 2 million arXiv research papers, including titles, abstracts, and categories.

- **Text Cleaning:** A custom text normalization function was used on both the abstract and title fields. This function eliminated excessive whitespace, removed non-alphanumeric characters (with the exception of punctuation that is important to sentence structure), and normalized text to lowercase for consistency.
- **Missing Value Removal:** To maintain the integrity and quality of the data, records that had null values in the abstract or title fields were not included.
- **Train-Test Split:** For reproducibility, the dataset was split using an 80/20 split with a fixed random seed into training and testing subsets.

V. Exploratory Data Analysis (EDA) and Hypotheses for the Study

A. Exploratory Data Analysis

EDA was primarily on arXiv metadata and used PyMuPDF to pull full-text PDFs for examining structural and semantic attributes of research articles.

The work examines both average length of abstract and distribution across the sample of 1000 research records. Chunking strategies were experimented to see if they are compatible with transformer models.

Readability and syntactic complexity of the documents were verified to establish if general-purpose and domain-specific models (such as PEGASUS, T5, BigBird) could generalize over scientific styles of language.

The study revealed frequency statistics of scientific items, patterns of keyword co-occurrence (e.g., "transformer", "language model", "architecture"), and quality of summaries generated by different models based on ROUGE-L scoring.

B. Hypotheses

- **The recently developed transformer-based models such as FLAN-T5, PEGASUS, and BART produce semantically richer and more contextual summaries of scientific documents than the less advanced or base summarization models (e.g., t5-small, distilbart).**
- **The RAG system, through FAISS-based retrieval and Sentence-BERT embeddings, surpasses standalone QA models in generating accurate and contextually appropriate responses by grounding responses in semantically significant document segments.**
- **Commercial LLMs (such as ChatGPT 3.5 Turbo) perform exceptionally well in complex QA scenarios compared to open-source versions on the basis of more extensive training data as well as upgraded generalization abilities. However, RAG systems offer more fact grounding if paired with accurate retrieved context.**

VI. Data Analytics

The end-to-end data analytics pipeline has several key steps, including both open-source and commercial LLM-based pipelines for question answering and summarization:

- **Summarization Models:** In total, 20 transformer-based summarization models were contrasted, e.g., BART, PEGASUS, T5, BigBird, and domain-specific variants like `pegasus-pubmed`. Three criteria informed model choice: transformer architecture diversity, academic domain relevance, and past performance on summarization benchmarks (ROUGE-L).
- **Question Generation:** Questions were generated using models like `flan-t5-base` and `pegasus-xsum`, which took summaries as input. This stage evaluated the models' ability to generate coherent, context-dependent questions based on academic text.
- **Answer Generation:** Three open-source models `flan-t5-large`, `flan-t5-xl`, and `bart-large-cnn-samsum`—produced the responses, while a RAG system built with FAISS and `declare-lab/flan-alpaca-base` was used for retrieval-augmented QA. Commercial LLMs like ChatGPT Turbo 3.5 were also evaluated for qualitative comparison.
- **Retrieval-Augmented Generation (RAG):** Dense retrieval was performed with Sentence-BERT embeddings stored with FAISS. Document chunks retrieved were appended with user queries to form context-sensitive prompts to the `flan-alpaca-base` model.
- **Evaluation Metrics:** The output quality was evaluated using ROUGE-L, BLEU, and cosine similarity. Based on this evaluation, comparisons of QA and RAG responses were done in terms of n-gram overlap, semantic similarity, and contextual accuracy.

VII. Data Visualization and Results Report

A. Research Design Workflow

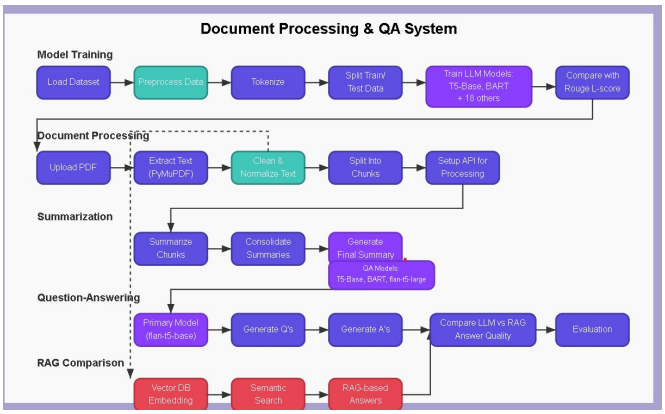

Fig. 1. Research Design Workflow

Workflow Explanation

This project compares the performance of transformer-based models in summarization, question generation, and question answering using a controlled and modular process. RAG pipelines are compared against paid LLMs (e.g., ChatGPT 3.5 Turbo) and their open-source alternatives.

1. Data Collection and Preprocessing

Datasets Used:

- *Summarization:* Kaggle's arXiv metadata (titles and abstracts).
- *PDF Content:* Actual scholarly articles that uploaded by the users and processed using PyMuPDF.

Preprocessing Steps:

- Text extraction with PyMuPDF. Token normalizing, stripping punctuation, lowercasing. Paragraph and chunk split for transformer input appropriateness (e.g., 300–500 tokens).

These processes deliver a clean, contextually tractable dataset for downstream uses like summarization and question generation.

2. Summarization Phase

Task: Evaluate 20 pre-trained summarization models (e.g., BART, PEGASUS, T5, BigBird, FLAN-T5, etc.).

Implementation: Academic text chunks are summarized using Hugging Face transformer models. Multi-step chunk-wise summarization is used for longer documents. Catch key information in semantically accurate summaries with a token overload reduction. Stored summaries as input for question generation models and then for evaluation.

3. Question Generation

Models Used: `flan-t5-base`, `pegasus-xsum`, and GPT-3.5 Turbo (via API).

Generate context-specific, related questions from abstracts with transformer-based generative models. Measure the capacity of models to extract implicit and explicit information to create legitimate questions. Fine-tuned prompts were required for GPT-3.5 to generate high-quality, multi-dimensional questions.

4. Question Answering Phase

Models Used:

- *Open-source QA:* `flan-t5-large`, `flan-t5-xl`, `bart-large-cnn-samsum`
- *RAG System:* `flan-alpaca-base` with FAISS and Sentence-BERT (`all-MiniLM-L6-v2`)
- *Paid LLM:* ChatGPT 3.5 Turbo

Workflow: FAISS indexes chunk sentence embeddings for semantic retrieval.Prompt (context + question) both combined and provided to generative QA models. QA answers are compared between RAG, open-source LLMs, and paid models.

5. Evaluation Metrics

Each task was evaluated using standard NLP metrics: ROUGE-1, ROUGE-2, ROUGE-L, BLEU, , F1 Score

Similarity Metrics: Cosine similarity (TF-IDF), ROUGE-L, BLEU, and semantic overlap using BERTScore

6. Result Comparison

The evaluation approach listed below was used to compare model results between QA pairs:

- Cosine Similarity (TF-IDF): Measures lexical similarity.
- ROUGE-L Score: Assesses longest sequence matches.

Similarity Comparison Between Open Source LLMs QA and RAG Answers:

- Cosine Similarity (0.176)
- ROUGE-L Score (0.162)

Similarity Comparison Between Commercial LLM QA and RAG Answers:

- Cosine Similarity (0.5157)
- ROUGE-L Score (0.476)

ROUGE-L Comparison for Summarization Models

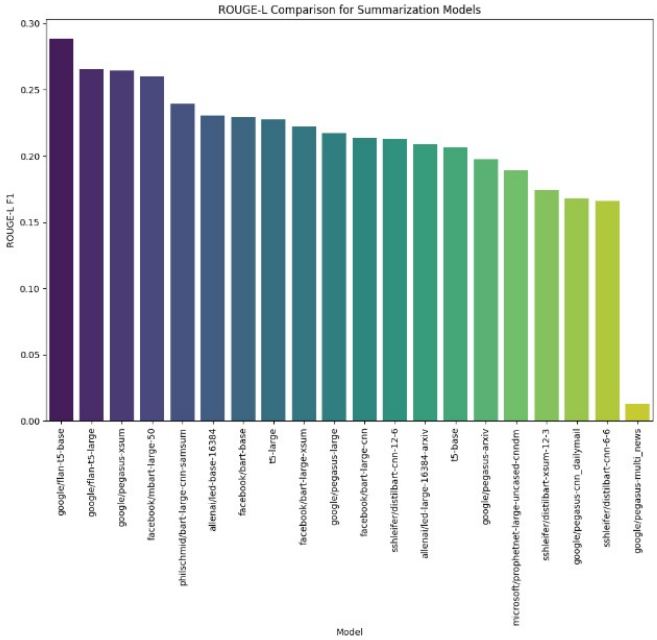

Fig. 2. ROUGE-L Comparison for Summarization Models

The top N words in the summary Generated by openai/gpt-3.5-turbo

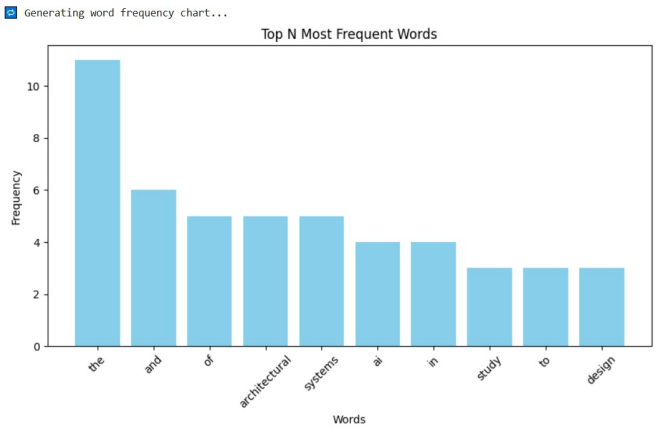

Fig. 3. The top N words in the summary

VIII. Conclusion

For overall summaries and straightforward questions, open-source models such as `flan-t5-xl`, `bart-large-cnn-samsum`, and `pegasus-xsum` performed well, with coherent and relevant outputs. In the case of technical or research-oriented questions, these models did not have the contextual depth, specificity, and accuracy required for good responses.

On the other hand, commercial LLMs, specifically ChatGPT (GPT-3.5 Turbo), consistently gave more precise, comprehensive, and contextually appropriate answers. These models demonstrated superior reasoning capabilities and a deeper grasp of complex scientific content.

Furthermore, cosine similarity comparisons between RAG system responses and commercial LLM responses revealed that while both approaches yield semantically dense output, commercial LLMs were more consistent in understanding and generating intricate academic material. This suggests the value of proprietary models for high-stakes or knowledge-heavy NLP tasks.

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